

AI-Scheduling-Assistant: Agentic Calendar Automation with LLMs

Arnav Kapoor (23060)

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Abstract

This project presents an intelligent scheduling assistant that leverages large language models (LLMs), Google Calendar APIs, and Hugging Face NLP pipelines to automate meeting coordination. The solution is designed for robust, autonomous operation, minimizing human intervention in scheduling, rescheduling, and optimizing meetings across multiple calendars. The folder has been reorganized for clarity and maintainability, and Hugging Face integration enables advanced natural language understanding for event analysis.

1 Introduction

Modern organizations face significant challenges in coordinating meetings, especially across distributed teams and busy calendars. Traditional tools require manual input and lack the intelligence to resolve conflicts or adapt to user preferences. The AI-Scheduling-Assistant aims to address these limitations by combining agentic AI principles with state-of-the-art LLMs and cloud APIs.

Motivation

- **Autonomy:** Reduce manual scheduling effort.
- **Accuracy:** Minimize conflicts and errors.
- **User Experience:** Enable natural language interaction and rapid response.

2 System Architecture

Folder Structure

- `src/`: Core Python scripts for event extraction, agent logic, and utilities.
- `notebooks/`: Jupyter notebooks for experiments, profiling, and demonstrations.
- `docs/`: Documentation and markdown guides.
- `samples/`: Example input/output JSON files.
- `tests/`: Test scripts and validation notebooks.

Key Components

- **Google Calendar Integration:** Secure authentication and event retrieval using OAuth2 and Google Calendar API.
- **Agentic AI Logic:** Python classes and functions for parsing meeting requests, extracting constraints, and proposing optimal times.
- **LLM Inference:** Integration with vLLM server and OpenAI API for natural language understanding.
- **Hugging Face NLP:** Sentiment analysis and text classification on event summaries using transformer models.

3 Workflow

1. Authentication: User credentials are loaded from token files for secure API access.
2. Event Extraction: Calendar events are fetched for specified users and date ranges.
3. Data Processing: Events are parsed to extract start/end times, attendees, and summaries.
4. NLP Enhancement: Hugging Face transformers analyze event summaries for sentiment and intent.
5. Scheduling Logic: The agent proposes meeting times, resolves conflicts, and adapts to user constraints.
6. Output: Results are structured in standardized JSON formats for downstream consumption.

4 Hugging Face Integration

A new notebook cell demonstrates Hugging Face's `transformers` pipeline for sentiment analysis:

```
from transformers import pipeline
nlp = pipeline('sentiment-analysis')
for e in event:
    summary = e.get('Summary', '')
    if summary:
        sentiment = nlp(summary)[0]
        print(f"Summary: {summary}\nSentiment: {sentiment['label']} (score: {sentiment['score']})")
```

This enables deeper insights into meeting topics, helping the agent prioritize or flag sensitive events.

5 Documentation

- **README.md:** Comprehensive guide to setup, usage, and architecture.
- **GoogleAuth.md:** Instructions for credential management.
- **Sample Notebooks:** Step-by-step demonstrations for event extraction, agent setup, and profiling.

6 Evaluation & Results

- **Autonomy:** The agent can schedule meetings with minimal human input.
- **Scalability:** Supports multiple users and calendars.
- **Extensibility:** Modular design allows easy integration of new models or APIs.
- **NLP Performance:** Hugging Face models provide reliable sentiment and intent detection.

7 Conclusion & Future Work

The AI-Scheduling-Assistant demonstrates the power of agentic AI and LLMs in automating complex scheduling tasks. Future enhancements may include:

- Advanced intent extraction (e.g., topic modeling, urgency detection).
- Integration with additional calendar providers.
- Real-time conflict resolution and user feedback loops.

8 References

- Google Calendar API Documentation
- Hugging Face Transformers Library
- vLLM: High-throughput LLM inference
- OpenAI API Documentation